



FROSTMODE — Project Technical Report & Documentation

"LOCK IN. LEVEL UP."

Full-Stack Winter Arc Daily Routine & Habit Tracking Web Application



Executive Project Summary

FrostMode is a full-stack gamified web application built for the **Winter Arc 90-day discipline challenge**. It helps users lock in, execute daily routines, track progress through **1,000 pre-loaded tasks**, enforce **Strict Mode penalties**, earn points, redeem rewards, and compete on a real-time global leaderboard.

- **Developer / Student Name:** Aayans Raj
- **Project Name:** FrostMode (Winter Arc Routine Tracker)
- **Live Netlify URL:** <https://frostmodewinterarc.netlify.app>
- **GitHub Repository:** <https://github.com/aayansraj/FrostMode>
- **Database:** Supabase Cloud PostgreSQL (1,000 Tasks)
- **Status:** 100% Complete, Deployed & Google Search Verified



Complete Technology Stack & Architecture

System Layer	Technology / Framework	Purpose & Technical Details
Frontend Framework	React 18 (Vite)	High-performance Single Page Application (SPA) built with modern component architecture.
Styling & Theme	Vanilla CSS + TailwindCSS	Dark winter glassmorphic aesthetic with custom CSS tokens, frosted panels, and glowing borders.
Animations & FX	HTML5 Canvas + Lucide React	Interactive custom ice cursor with dynamic trailing particles and 60 FPS falling snow background.
Backend API	Node.js + Express.js	Express server hosting REST endpoints for Auth, Tasks, Leaderboard, Analytics, and Rewards.
Real-Time Engine	Socket.io WebSockets	Real-time WebSocket connection for instant global leaderboard rankings and Top 3 podium updates.
Cloud Database	Supabase Cloud PostgreSQL	Cloud SQL database hosting 1,000 tasks across 7 relational schemas with AWS connection pooling.
Authentication	Google OAuth 2.0 + JWT + Bcrypt	Dual authentication: Google Identity Services + password-based auth with fail-proof session tokens.
Hosting & Deployment	Netlify + GitHub CI/CD	Continuous integration & deployment workflow from GitHub main branch to Netlify production.
SEO & Indexing	Google Search Console	Added google-site-verification meta tag, OpenGraph tags, ownership verification, and priority search indexing.

Step-by-Step Implementation Journey (What We Built)

1. Brand Identity & Animated SVG Logos

- Designed custom animated brand logos (`frostmode-logo-animated-dark.svg` , `frostmode-logo-animated-light.svg` , and `frostmode-icon-animated.svg`).
- Integrated CSS keyframe animations for glowing snowflake pulses and sleek typography matching the "Lock In. Level Up." theme.

2. Custom Interactive Ice Cursor & Snow FX

- Built `CustomCursor.jsx` with dynamic ice trail particle physics following user pointer movements.
- Created `SnowBackground.jsx` utilizing HTML5 Canvas rendering for a smooth winter snow effect.

3. Database Architecture & 1,000 Tasks Library

- Engineered a master JSON dataset (`winter_arc_tasks.json`) containing **1,000 Winter Arc tasks** categorized into **Health**, **Fitness**, **Mindset**, and **Productivity**.
- Created 7 database schemas:
 1. `users` — User profiles, level ranks, wallet points, streaks, avatars, and leaderboard visibility.
 2. `arcs` — 90-day Winter Arc setup, target days, and strict mode parameters.
 3. `days` — Daily routine tracking logs and completion notes.
 4. `library_tasks` — Master repository of all 1,000 tasks.
 5. `tasks` — User active daily task commitments.
 6. `points_logs` — Transaction history of earned points and strict mode penalties.
 7. `rewards` — Custom store reward redemptions.

4. Supabase Cloud PostgreSQL Migration & Seeding

- Connected the backend server to **Supabase Cloud PostgreSQL** (`aws-0-ap-southeast-1.pooler.supabase.com:6543/postgres`).
- Built an automated batch seeder (`initPostgres.js`) that creates database tables and inserts all **1,000 tasks** and demo leaderboard warriors in under 1 second.

5. Google OAuth 2.0 & Fail-Proof Authentication

- Integrated Google OAuth Client ID `1061825611076-r9st2ddp2vcv29a1m9m1m5r83h9ejdg6.apps.googleusercontent.com` .
- Added an eye toggle (`Eye` / `EyeOff`) for password visibility.
- Implemented `AuthContext.jsx` with fail-proof session handlers so that login, signup, and Google OAuth work seamlessly across both local and production environments.

6. Executive Glassmorphic Navbar Redesign

- Refactored `Navbar.jsx` into a floating glassmorphic container with single-line `whitespace-nowrap` labels (`Dashboard` , `Routine` , `Library` , `Leaderboard` , `Store` , `Analytics` , `Settings`).
- Added user level rank badge (`👶 Beginner`), snowflake points counter (`❄️ 0 pts`), strict mode badge (`🔥 STRICT`), theme toggle, and mobile drawer menu.

7. Netlify Production Deployment & Router Fixes

- Deployed site to Netlify URL `https://frostmodewinterarc.netlify.app` .

- Configured `netlify.toml` build commands (`npm install --prefix client && npm run build --prefix client`) and SPA redirects (`/* /index.html 200`) to eliminate 404 page refresh issues.

8. Google Search Indexing & Ownership Verification

- Added Google site verification tag (`YVrSh6zte2biHSoLdRdTngmIVQW4dXYizKE0obyDTYc`) and OpenGraph SEO tags in `client/index.html` .
- Successfully verified website ownership on **Google Search Console** and submitted a priority indexing request for Google Search results.

Live Credentials & Verification Links

-  **Live Website (Netlify Production):** <https://frostmodewinterarc.netlify.app>
-  **GitHub Source Code:** <https://github.com/aayansraj/FrostMode>
-  **Supabase Database Dashboard:**
<https://supabase.com/dashboard/project/iigurmvwjkjsnmhhgwgg>
-  **Local Development App:** <http://localhost:3000>

Report Prepared by: Aayans Raj

Project: FrostMode — Winter Arc Tracker

Tagline: "Lock In. Level Up."